

MODERN SLAVERY STATEMENT ANALYSIS

Statement Auditor

Scores public corporate modern-slavery statements on **disclosure quality** — how specific and substantiated they are — and catches when a company just re-files last year's.

Automated analysis of public statements — **not a legal assessment**. It measures statement quality and disclosure specificity. It never asserts that any company uses forced labour.

selimsevim.github.io/statement-auditor github.com/selimsevim/statement-auditor

The law requires a statement. It doesn't grade one.

- Every large UK company must publish a modern-slavery statement. **Thousands do** — the registry holds 100,000+.
- Nobody reads them at scale, and **the Act doesn't measure quality**.
- So a company can file **four pages of boilerplate**, or re-file last year's, and tick the box.

UK Modern Slavery Act 2015 — s.54(5)

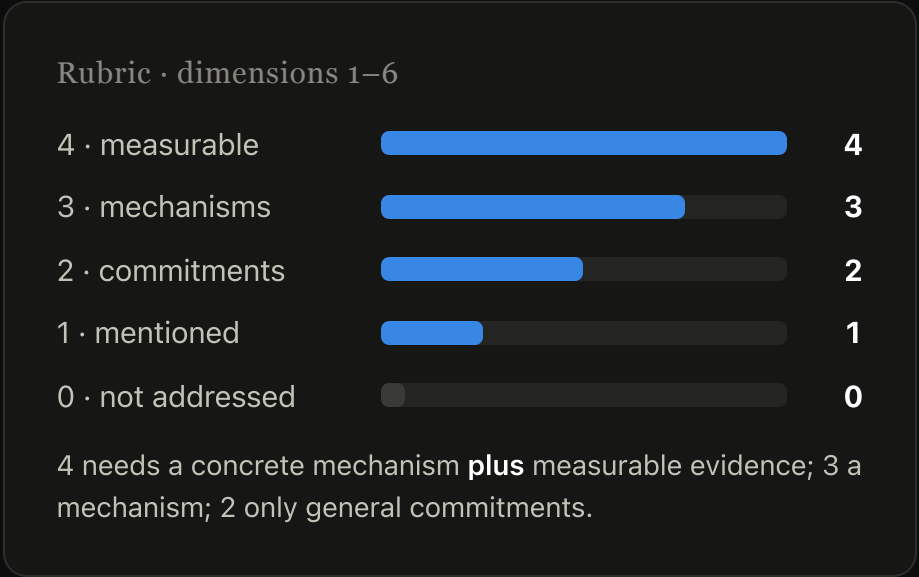
Six recommended content areas

Structure & supply chains · Policies · Due diligence · Risk assessment · Effectiveness & KPIs · Training.

The Act **recommends** what to disclose but sets **no bar** for how well. That gap is what we score.

Seven dimensions, anchored to the Act

- 1–6 map directly to s.54(5)'s six content areas, each scored 0–4 against our rubric.
- 7 · Boilerplate is **computed**, not LLM-scored — year-over-year paragraph reuse.
- Overall = mean of dimensions 1–6, **penalised by boilerplate**.
- If >60% of paragraphs are carried over, the overall is **capped at 2.0** and flagged.



Two LLM passes, never one giant prompt

PASS 1 · EXTRACT

Claims

Claude Haiku 4.5 pulls verbatim claims per dimension into strict JSON. PDFs read column-aware with PyMuPDF so quotes stay intact.

PASS 2 · SCORE

Rubric 0–4

Claude Sonnet 5 scores each dimension, sampled **3×** with the median taken. Every cited quote is **verified verbatim** in the source — or dropped.

COMPUTED

Boilerplate diff

Paragraph embeddings (**all-MiniLM-L6-v2**), year-over-year cosine similarity at a **0.95** calibrated threshold, plus hedge-word density.

All LLM responses are cached on **(model, prompt hash)** — re-runs are free and the committed snapshot is deterministic.

11 statements · 7 companies · scored 1.5 → 3.5

Canon Europe · 2022		3.5
Tesco PLC · 2024		3.5
Barclays PLC · 2024		3.3
Dell Technologies · 2022		3.2
Damen Shipyards · 2022		2.2
Dell Technologies · 2023 ○		2.0
Siemens Gamesa · 2022		1.5

11

STATEMENTS

7

COMPANIES

6

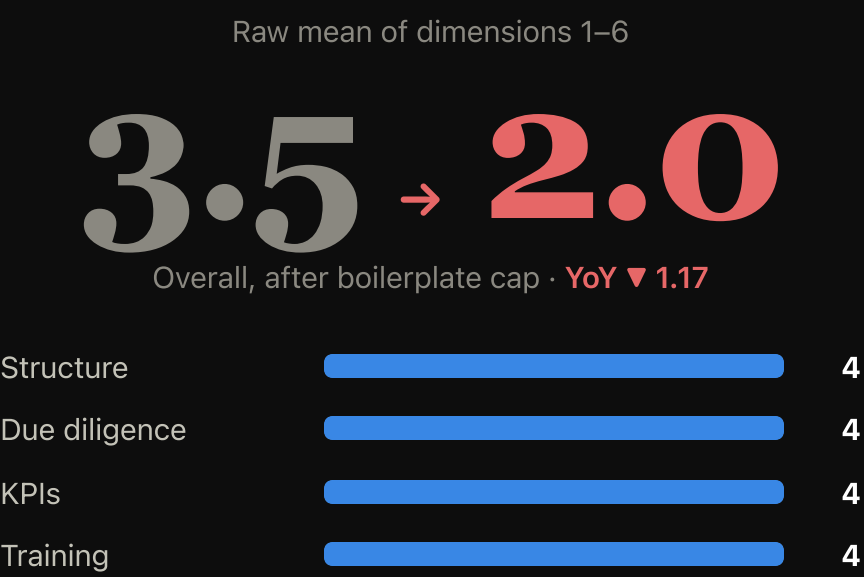
YOY PAIRS

1

CAPPED

Sampled from the UK government's official **Modern Slavery Registry** CSV export — consecutive-year pairs, not hand-picked. ○ red ring = boilerplate-capped.

Dell 2023: well-written, and recycled



70% carried over from 2022

39 of 56 paragraphs are substantially identical to the prior year. They re-filed the document and changed a few numbers.

This isn't a guess: every paragraph is embedded and must clear **95% cosine similarity** — a threshold calibrated by hand against real pairs. Above 60% carried over, the overall is capped at 2.0.

That's the gap the raw score would hide — and exactly what box-ticking compliance misses.

Every score cites the source's own words

- Each dimension expands to its **justification and verbatim evidence** — the exact span sliced from the PDF text.
- The scoring model **can't invent a quote**: if a cited quote doesn't string-match the source, we drop it.
- If a dimension loses all its evidence, it's **scored 0**. No hallucinated praise.
- The dashboard highlights by **stored character offset** — never re-matched in the browser.

"...verified to appear verbatim in the extracted source text, highlighted by the character offset stored at scoring time."

✓ source-verified

quote dropped if unmatched

no evidence → 0

A reviewer opening the original PDF finds the exact sentence — cosmetic differences (case, bullets, hyphenation) tolerated, wording never paraphrased.

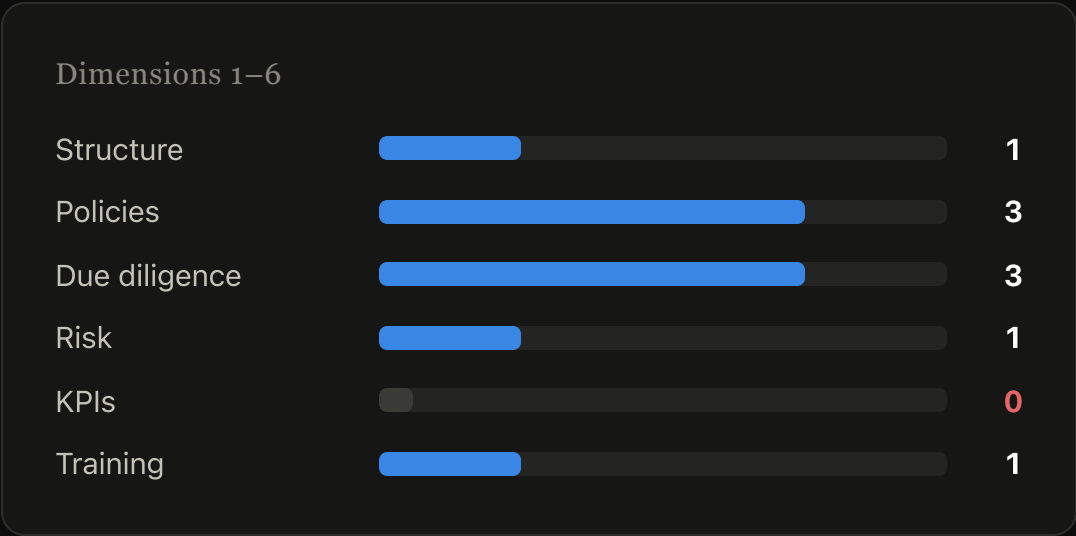
Siemens Gamesa 2022 — a clean 1.5

A multi-billion-euro wind-turbine maker, four-page statement. It has **no measurable KPIs at all** — a genuine zero, not a pipeline miss.

We're honest about our own pipeline: **fallback** tags mean a dimension had no extracted claims, so we re-read the whole statement to avoid a false zero — **capped at 2**, because a passing mention isn't a mechanism.

fallback · risk

fallback · training



THE ACCOUNTABILITY LAYER THAT'S MISSING

Grounded in the law. Every score backed by evidence.

Grounded

Rubric anchored to UK MSA s.54(5) — not an invented checklist.

Verifiable

Every score cites a verbatim, source-matched quote — or scores 0.

Catches recycling

Surfaces the boilerplate that box-ticking compliance misses.

The registry has **100,000+ statements**. This scores any of them — per-statement and cache-backed.

Live demo → selimsevim.github.io/statement-auditor github.com/selimsevim/statement-auditor